Games — problem 13.13

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A crosses out any 2^7 of the numbers $0, 1, \ldots, 255, 256$. Then B crosses out any 2^6 numbers. Then A crosses out any 2^5 numbers, and so on until finally B crosses out $2^0 = 1$ number. Since $2^7 + 2^6 + \cdots + 2^0 = 2^8 - 1$ numbers are crossed out, there will be two numbers a and b left. B pays the difference $|a-b|$ to A. How should A play to get as much as possible ? How should B play to lose as little as possible ? How much does A win per game if both players use their optimal strategies ?